

Poisson Regression

Last time: Another dataset

A concerned parent asks us to investigate crime rates on college campuses. We have access to data on 81 different colleges and universities in the US, including the following variables:

- + type: college (C) or university (U)
- + nv: the number of crimes for that institution in the given year
- + enroll1000: the number of enrolled students, in thousands
- + region: region of the US C = Central, MW = Midwest, NE = Northeast, SE = Southeast, SW = Southwest, and W = West)

Last time

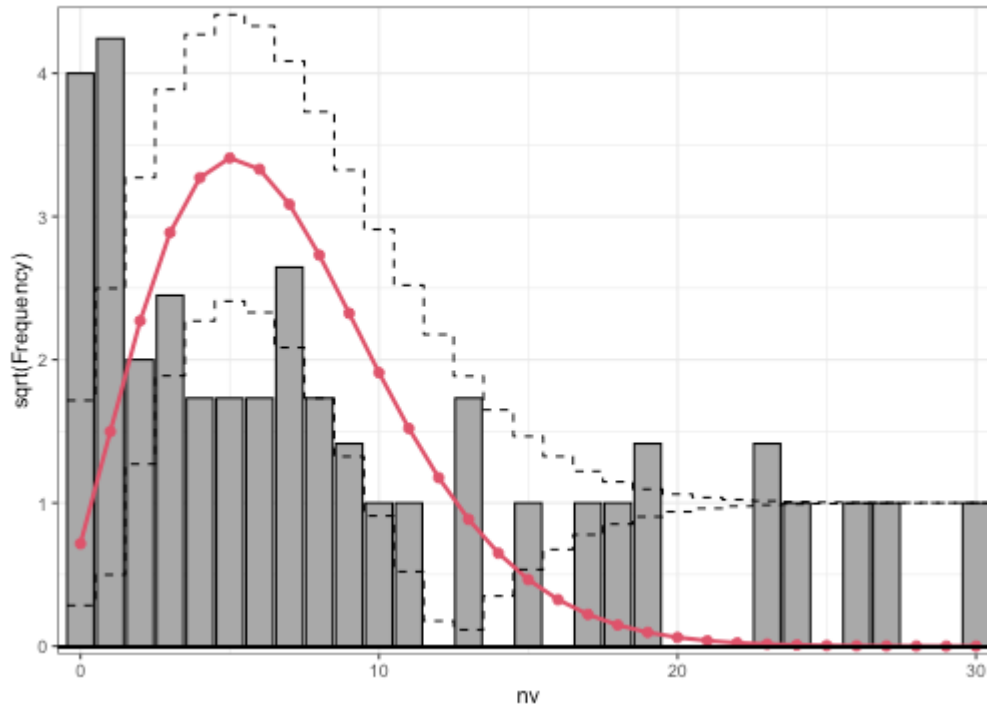
We want to know whether there are regional differences in the number of crimes on college campuses.

$$Crimes_i \sim Poisson(\lambda_i)$$

$$\log(\lambda_i) = \beta_0 + \beta_1 MW_i + \beta_2 NE_i + \beta_3 SE_i + \beta_4 SW_i + \beta_5 W_i$$

Last time: Rootogram

```
m1 <- glm(nv ~ region, data = crimes, family = poisson)
rootogram(m1, style="standing")
```



Goodness of fit

Another way to assess whether our model is reasonable is with a *goodness of fit* test.

Goodness of fit test: If the model is a good fit for the data, then the residual deviance follows a χ^2 distribution with the same degrees of freedom as the residual deviance

...

```
## Residual deviance: 621.24    on 75    degrees of freedom
```

```
## AIC: 851.12
```

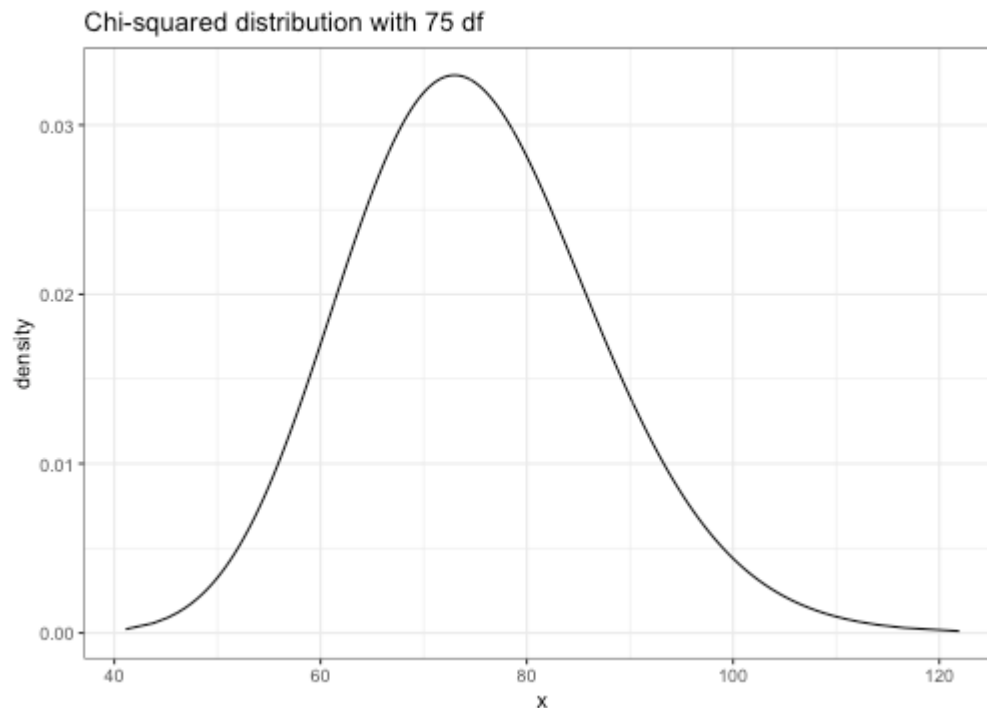
...

Residual deviance = 621.24, df = 75

How likely is a residual deviance of 621.24 if our model is correct?

Goodness of fit

Residual deviance = 621.24, df = 75



```
pchisq(621.24, df=75, lower.tail=F)
```

```
## [1] 5.844298e-87
```

Goodness of fit

Goodness of fit test: If the model is a good fit for the data, then the residual deviance follows a χ^2 distribution with the same degrees of freedom as the residual deviance

Residual deviance = 621.24, df = 75

```
pchisq(621.24, df=75, lower.tail=F)
```

```
## [1] 5.844298e-87
```

So our model might not be a very good fit to the data.

Potential issues with our model

- + The Poisson distribution might not be a good choice
- + There may be additional factors related to the number of crimes which we are not including in the model

Offsets

We will account for school size by including an **offset** in the model:

$$\log(\lambda_i) = \beta_0 + \beta_1 MW_i + \beta_2 NE_i + \beta_3 SE_i + \beta_4 SW_i + \beta_5 W_i \\ + \log(Enrollment_i)$$

Motivation for offsets

We can rewrite our regression model with the offset:

$$\log(\lambda_i) = \beta_0 + \beta_1 MW_i + \beta_2 NE_i + \beta_3 SE_i + \beta_4 SW_i + \beta_5 W_i \\ + \log(Enrollment_i)$$

Fitting a model with an offset

```
m2 <- glm(nv ~ region, offset = log(enroll1000),
          data = crimes, family = poisson)
summary(m2)
```

```
...
## regionSE      0.87237      0.15313      5.697 1.22e-08 ***
## regionSW      0.50708      0.18507      2.740 0.00615 **
## regionW       0.20934      0.18605      1.125 0.26053
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
##
## (Dispersion parameter for poisson family taken to be 1)
...
```

- ✚ The offset doesn't show up in the output (because we're not estimating a coefficient for it)

Fitting a model with an offset

$$\begin{aligned}\log(\hat{\lambda}_i) = & -1.30 + 0.10MW_i + 0.76NE_i + \\ & 0.87SE_i + 0.51SW_i + 0.21W_i \\ & + \log(Enrollment_i)\end{aligned}$$

How would I interpret the intercept -1.30?

When to use offsets

Offsets are useful in Poisson regression when our counts come from groups of very different sizes (e.g., different numbers of students on a college campus). The offset lets us interpret model coefficients in terms of rates instead of raw counts.

With your neighbor, brainstorm some other data scenarios where our response is a count variable, and an offset would be useful. What would our offset be?

Inference

$$\log(\lambda_i) = \beta_0 + \beta_1 MW_i + \beta_2 NE_i + \beta_3 SE_i + \beta_4 SW_i + \beta_5 W_i \\ + \log(Enrollment_i)$$

Our concerned parent wants to know whether the crime rate on campuses is different in different regions.

What hypotheses would we test to answer this question?

Drop-in-deviance test

Full model:

$$\log(\lambda_i) = \beta_0 + \beta_1 MW_i + \beta_2 NE_i + \beta_3 SE_i + \beta_4 SW_i + \beta_5 W_i \\ + \log(Enrollment_i)$$

Reduced model:

$$\log(\lambda_i) = \beta_0 + \log(Enrollment_i)$$

Drop-in-deviance test

```
m2 <- glm(nv ~ region, offset = log(enroll1000),  
          data = crimes, family = poisson)  
summary(m2)
```

...

```
##      Null deviance: 491.00  on 80  degrees of freedom
```

```
## Residual deviance: 433.14  on 75  degrees of freedom
```

```
## AIC: 663.02
```

```
##
```

...

What is my test statistic?

Drop-in-deviance test

```
m2 <- glm(nv ~ region, offset = log(enroll1000),  
          data = crimes, family = poisson)  
summary(m2)
```

...

```
##      Null deviance: 491.00   on 80   degrees of freedom  
## Residual deviance: 433.14   on 75   degrees of freedom  
## AIC: 663.02  
##
```

...

$$G = 491 - 433.14 = 57.86$$

```
pchisq(57.86, df=5, lower.tail=F)
```

```
## [1] 3.361742e-11
```

Inference

$$\log(\lambda_i) = \beta_0 + \beta_1 MW_i + \beta_2 NE_i + \beta_3 SE_i + \beta_4 SW_i + \beta_5 W_i \\ + \log(Enrollment_i)$$

Now our concerned parent wants to know about the difference between Western and Central schools. They would like a "reasonable range" of values for the difference between the regions.

How would we construct a "reasonable range" of values for this difference?

Confidence intervals

$$\log(\lambda_i) = \beta_0 + \beta_1 MW_i + \beta_2 NE_i + \beta_3 SE_i + \beta_4 SW_i + \beta_5 W_i \\ + \log(Enrollment_i)$$

...

```
## (Intercept) -1.30445      0.12403 -10.517    < 2e-16 ***
## regionMW      0.09754      0.17752   0.549    0.58270
## regionNE      0.76268      0.15292   4.987    6.12e-07 ***
## regionSE      0.87237      0.15313   5.697    1.22e-08 ***
## regionSW      0.50708      0.18507   2.740    0.00615 **
## regionW       0.20934      0.18605   1.125    0.26053
## ---
```

...

95% confidence interval for β_5 :

Class activity

Work on the class activity (handout).

Class activity

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.304617	0.102981	2.958	0.0031	**
femWomen	-0.224594	0.054613	-4.112	3.92e-05	***
marMarried	0.155243	0.061374	2.529	0.0114	*
kid5	-0.184883	0.040127	-4.607	4.08e-06	***
phd	0.012823	0.026397	0.486	0.6271	
ment	0.025543	0.002006	12.733	< 2e-16	***

We are interested in the relationship between prestige and the number of articles published, after accounting for other factors. What confidence interval should we make?

Class activity

$$Articles_i \sim Poisson(\lambda_i)$$

$$\log(\lambda_i) = \beta_0 + \beta_1 Female_i + \beta_2 Married_i + \beta_3 Kids_i + \beta_4 Prestige_i + \beta_5 Mentor_i$$

Do I need an offset for this model?